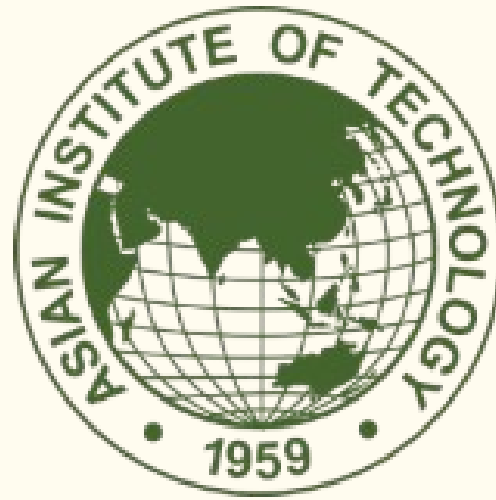




Lab for Week 6

Exploratory Data Analysis

Computer Programming for DS and AI

September 15, 2026

What are the key components of EDA?

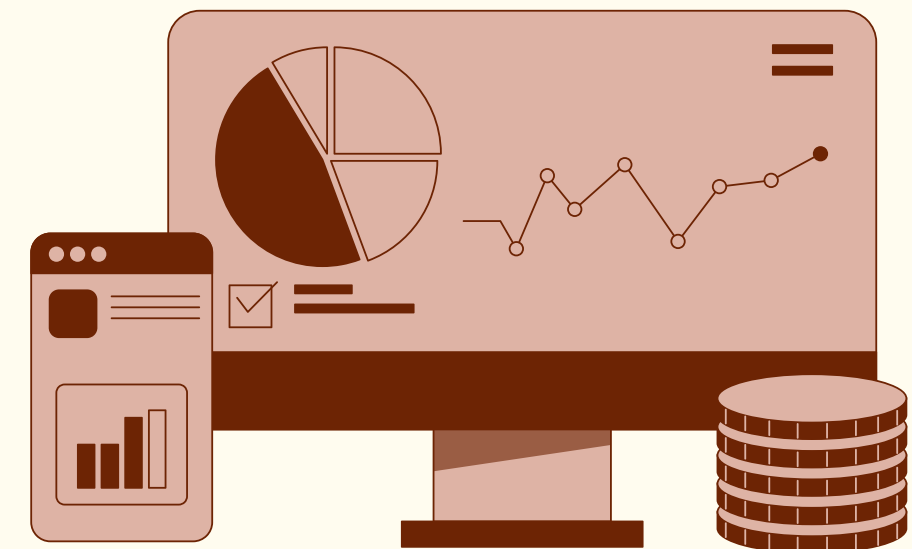


1. Data Loading & Inspection

- Loading data
- Exploring data (.head(), .shape, .columns)
- Structural summary (.info(), .dtypes)
- Statistical summary (.describe())

2. Data Cleaning

- Checking for duplicate rows
- Checking for missing values
- Imputing missing values
- Dropping irrelevant columns
- Dtype fixes



What are the key components of EDA?

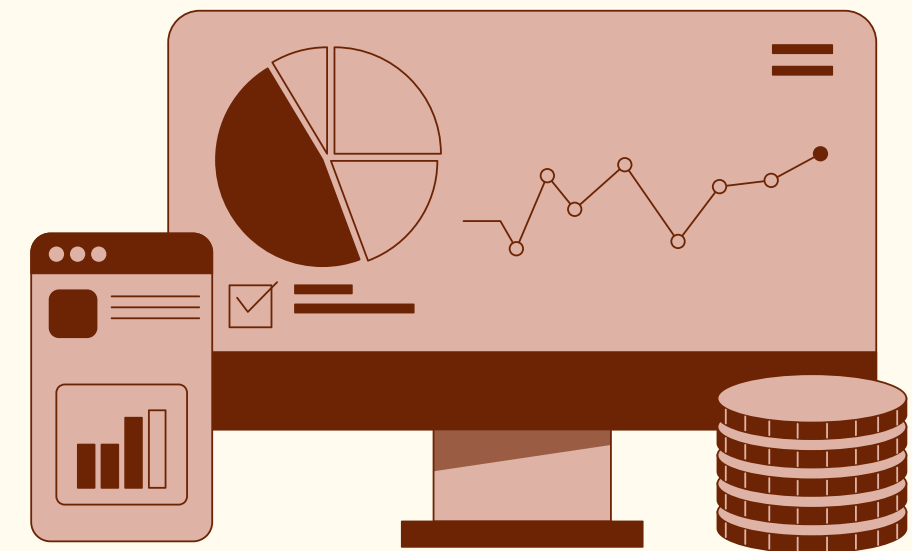


3. Encoding Categorical Variables

- Label Encoding (ordinal-style integer mapping)
- One-Hot Encoding

4. Binning

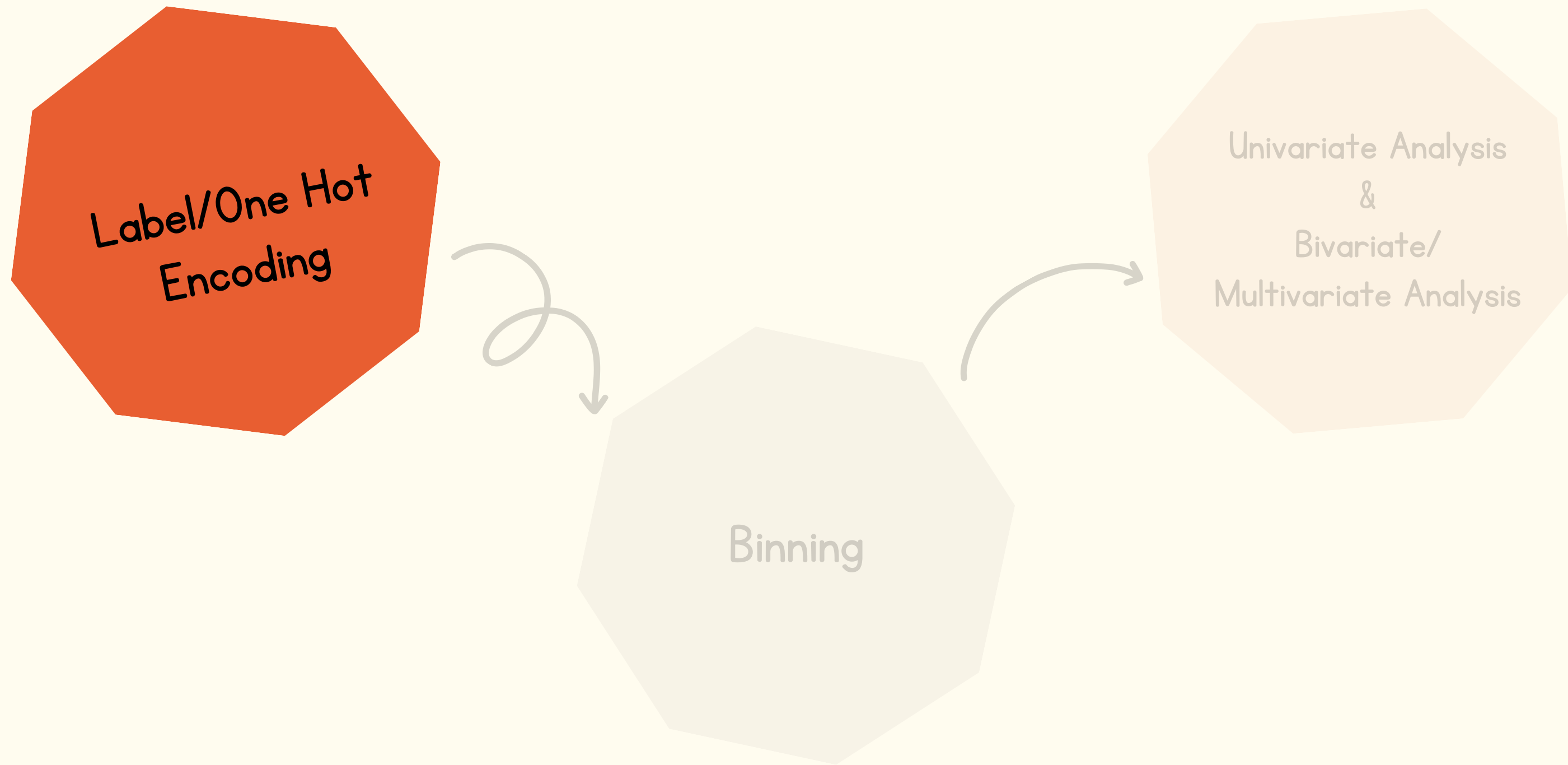
- Converting continuous variables into categorical buckets
- Choosing bin edges



5. Univariate Analysis

6. Bivariate / Multivariate Analysis





Label Encoding

Label Encoding: categorical value to numerical value

Original Data

Team	Points
A	25
A	12
B	15
B	14
B	19
B	23
C	25
C	29



Label Encoded Data

Team	Points
0	25
0	12
1	15
1	14
1	19
1	23
2	25
2	29

Encoding is typically assigned alphabetically:

A → 0

B → 1

C → 2

D → 3

Label Encoding (cont..)

Best suited when:

- The categorical variable has few unique categories (low cardinality)
- There's an order that genuinely exists in dataset. (e.g., "Low/Medium/High")

Be cautious when:

- Categories are purely nominal (no natural rank)
- The model may wrongly assume $2 > 1 > 0$ means something

One hot encoding

One-Hot Encoding: categorical value to numerical value, using a separate binary column per category

Original Data		One-Hot Encoded Data			
Team	Points				
A	25	1	0	0	25
A	12	1	0	0	12
B	15	0	1	0	15
B	14	0	1	0	14
B	19	0	1	0	19
B	23	0	1	0	23
C	25	0	0	1	25
C	29	0	0	1	29

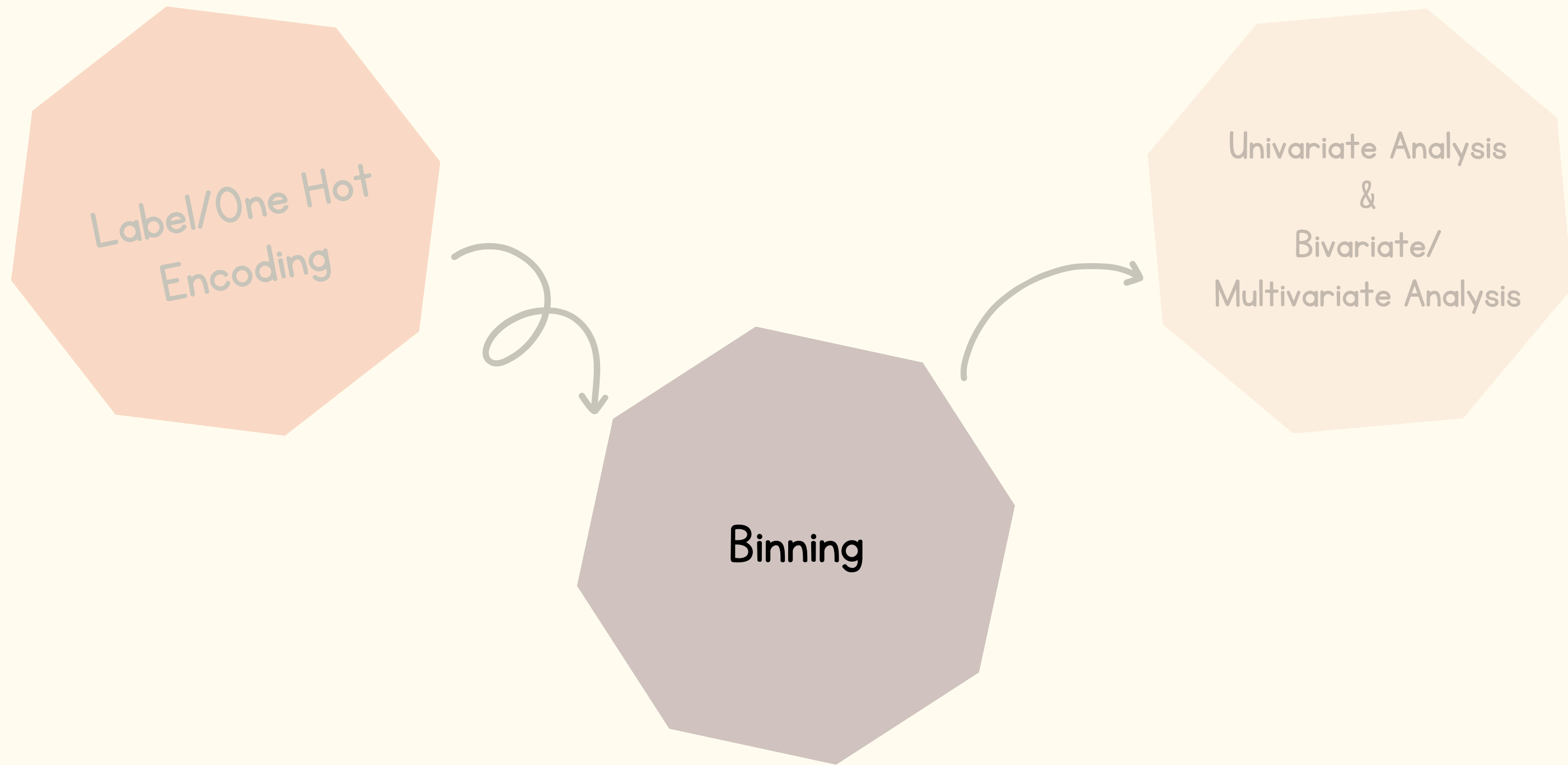
One hot encoding (cont...)

Best suited when:

- The categorical variable is nominal, no natural order between categories
- The variable has low-to-moderate cardinality (too many categories → too many new columns)
- You want to avoid implying a false numeric relationship between categories

Be cautious when:

- Cardinality is high (e.g., features with 200+ values). One-hot encoding would create an unmanageable number of sparse columns



Binning

Binning: numerical value to categorical value

Price (Original)	Price Category (Binned)
\$35	Budget
\$80	Mid-range
\$150	High
\$450	Luxury

Bins:

- 0-50 → Budget
- 50-100 → Mid-range
- 100-200 → High
- 200+ → Luxury

Binning (cont...)

Best suited when:

- A continuous variable is skewed or spans a huge range, making raw values hard to interpret visually
- You want to simplify a variable for grouped comparisons (e.g., countplots, groupby) rather than exact-value analysis

Be cautious when:

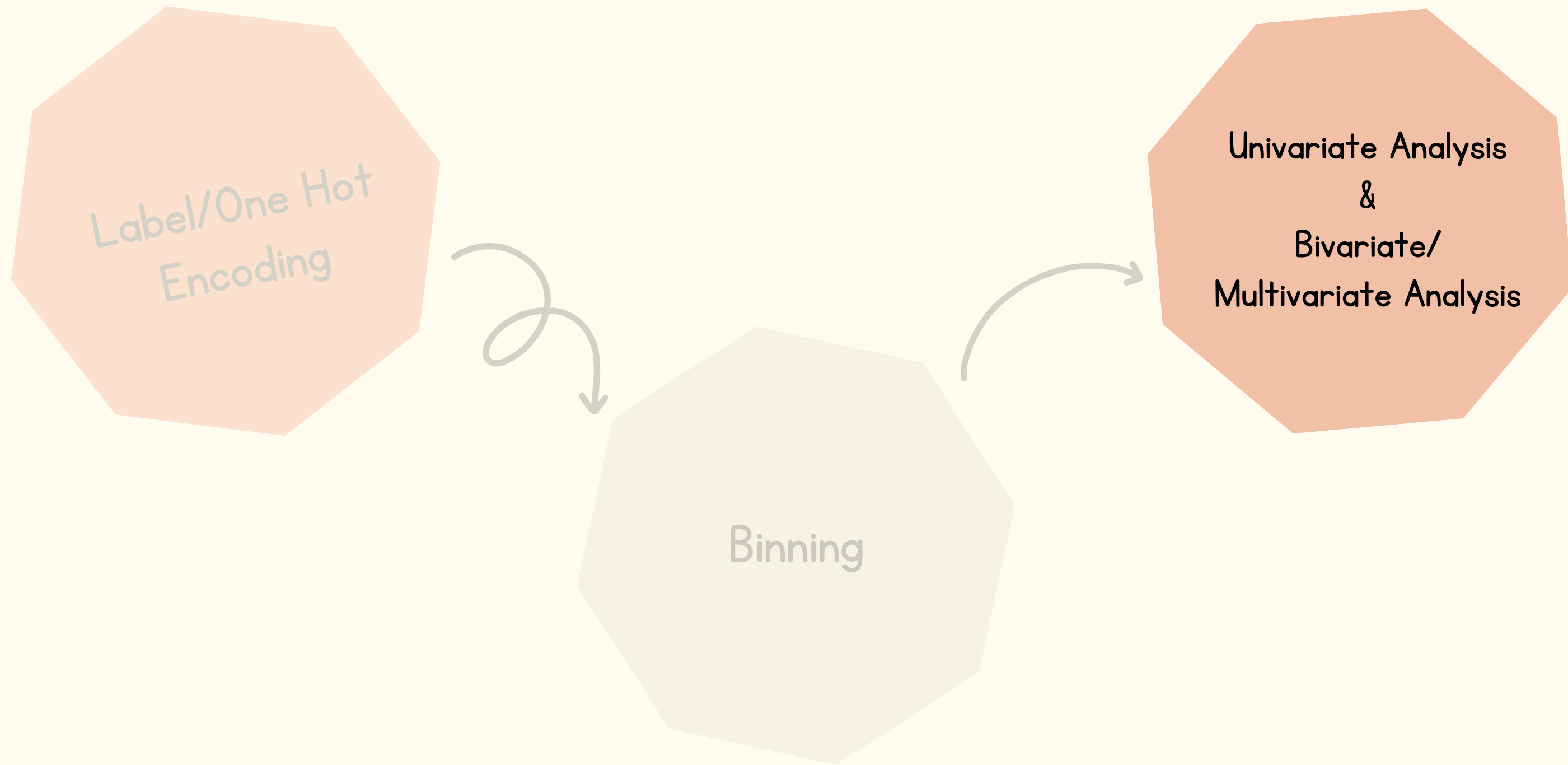
- Bin edges are chosen carelessly, arbitrary or evenly-spaced bins can hide real patterns if the data isn't evenly distributed
- Binning discards information, always a trade-off between simplicity and precision

Binning (cont...)

Choosing Bin Edges

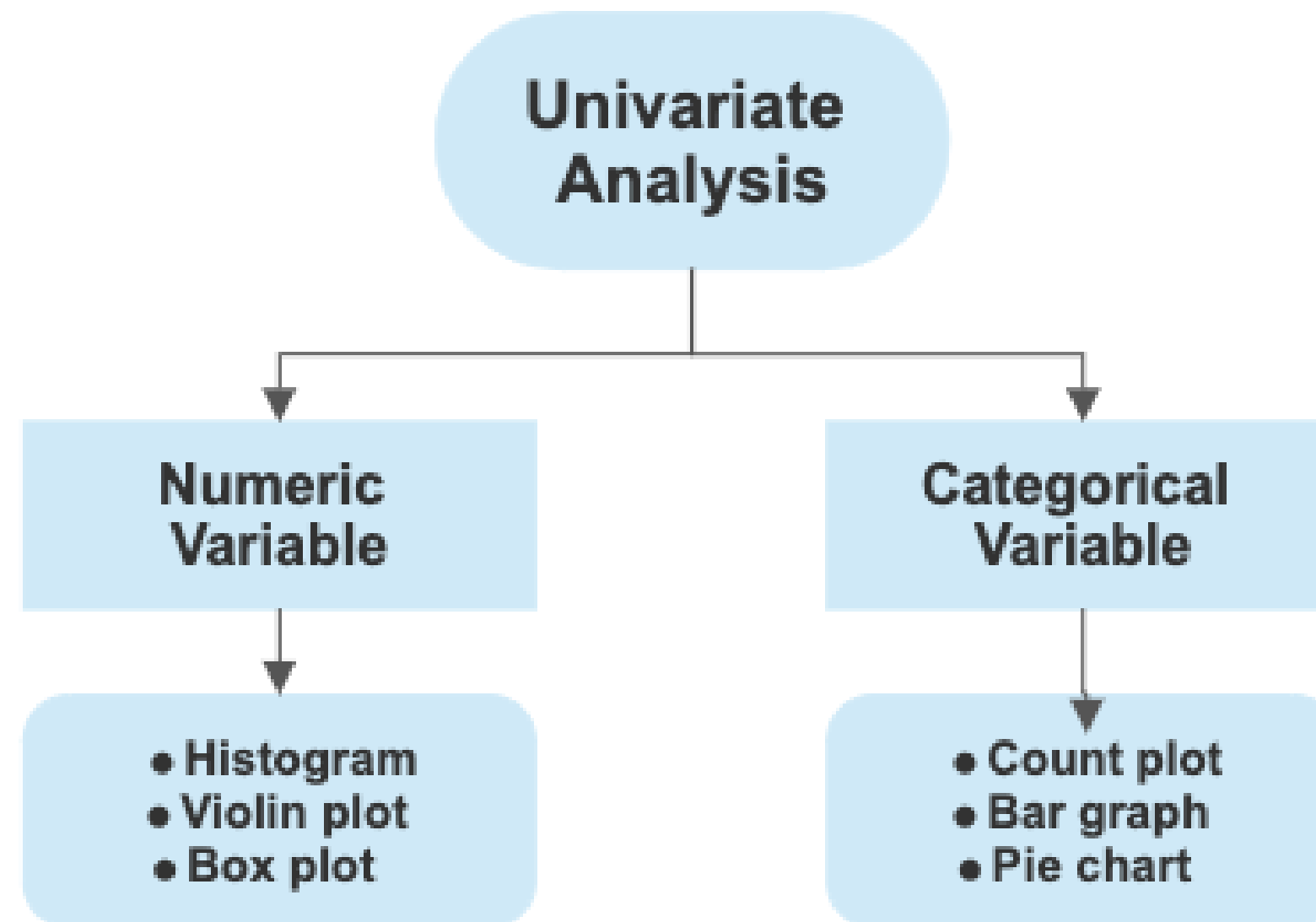
- **Quantile-based:** Use percentiles (25th, 50th, 75th) from `.describe()` so each bin has a roughly similar number of observations.
- **Domain knowledge:** Use meaningful real-world thresholds (e.g., tenure: 0-12 months = "New," 12-48 = "Established," 48+ = "Long-term")
- **Equal-width bins:** Split the range into equal-sized intervals (e.g., every \$50). Simple, but can create very uneven group sizes if the data is skewed. One bin might hold 90% of the data.

Avoid arbitrary guessing. Always check `.describe()` or a histogram first so your bin edges reflect where the data actually clusters, not a guess.



Univariate Analysis

Univariate Analysis: focus on one variable (feature) at a time



Bivariate/Multivariate Analysis

Bivariate/Multivariate Analysis: focus on relationships between two or more variables

